Cube-Sum Permutations: 60 Integer Identities and Five Algebraic Templates

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Abstract

Let $P(n)$ count permutations p of $\{1,\dots,n\}$ for which every $i+p(i)$ is a perfect cube. We prove $P(n)>0$ for every $n \geq 173$ from 60 exceptional integer endpoint identities (816 reflected blocks) and five general algebraic block-rearrangement templates giving sixteen further seeds. An elementary suffix-reflection rule and a cubic-gap inequality extend the construction to all later lengths. An explicit predecessor table explains all remaining finite lengths without a matching search, a large permutation archive, or dependent augmenting paths. The finite case distinctions are *not* eliminated: this is a solver-free, hand-checkable certificate rather than a case-free parameterized construction.

1 Two elementary lemmas

Lemma 1 (Boundary cancellation)**.** *Let $0=b_0<b_1<\cdots<b_t=n$ and let $c_1,\dots,c_t$ be positive integer cubes. Set $L_j=c_j-b_j$ and $R_j=c_j-b_{j-1}$. Assume $1\leq L_j<R_j\leq n+1$. Then $p(i)=c_j-i$ for $b_{j-1}<i\leq b_j$ is a cube-sum permutation if and only if*

$$\boxed{\{1,R_1,\dots,R_t\}_{\text{multi}}=\{n+1,L_1,\dots,L_t\}_{\text{multi}}.}$$

Proof. Each input block maps bijectively onto the integer interval $[L_j,R_j)$. At an integer x, the jump in the total interval-coverage count is the multiplicity of x as a left endpoint minus its multiplicity as a right endpoint. The displayed identity makes that jump 1 at $x=1$, -1 at $x=n+1$, and zero everywhere else. Thus coverage is exactly one at every $x\in[1,n]$. Reversing this argument proves necessity. □

Lemma 2 (Growing cubic suffix)**.** *Suppose $0\leq m<n$, $m+n+1=c$ is a cube, and a cube-sum permutation on $[1,m]$ exists (the $m=0$ permutation is empty). Keep the existing permutation on $[1,m]$ and set $p(i)=c-i$ for $m<i\leq n$. Then p is a cube-sum permutation on $[1,n]$.*

Proof. The new reflection is a bijection from $[m+1,n]$ to itself, and each new sum is c. □

2 Five algebraic block families and 60 exceptional identities

Lemma 3 (Reflected-block rearrangement)**.** *Choose positive input-block lengths $\ell_1,\dots,\ell_t$ of sum n and an arbitrary output order π of these t blocks. Let A_j be the sum of input lengths before block j, and let B_j be the sum of output lengths before block j in order π. If all numbers*

$$C_j=1+A_j+B_j+\ell_j$$

are perfect cubes, then on input block j the reflection $p(i)=C_j-i$ gives a cube-sum permutation of $[1,n]$.

Proof. The input block is $[A_j+1,A_j+\ell_j]$, which maps exactly to the prescribed output block $[B_j+1,B_j+\ell_j]$. Those output blocks partition $[1,n]$ by construction. Each index sum is C_j. □

Family I (four blocks). Let $U<V$ be integer cubes and put

$$a=2U-V-1,\qquad b=2V-3U+1.$$

If $a,b>0$, use input lengths (a,b,a,b) and output order $(3,4,1,2)$. Lemma 3 gives the cubic sums (U,V,U,V) and length $2(V-U)$. The pairs $(U,V)=(64,125),(125,216),(216,343)$ yield lengths $122,182,254$ respectively.

Family II (five blocks). For cubes $X < Y < Z$, define

$$a = 2Y - Z - 1, \quad b = X + 2Z - 4Y + 1, \quad c = Y - X.$$

If $a, b, c > 0$, the lengths (a, b, c, a, c) in output order $(4, 2, 5, 1, 3)$ produce sums (Y, X, Z, Y, Z) and total length $2Y - X - 1$. With $(Y, Z) = (64, 125)$ and $X = 27, 8$, the results are $n = 100, 119$.

Family III (six blocks). For integer cubes X, W, Y, Z, put

$$\begin{aligned} a &= 2Y - Z - 1, & b &= X - 2a - 1, \\ c &= W + 2a - 2X + 1, & d &= Y + X - W - 2a - 1. \end{aligned}$$

If these are positive, choose lengths (a, b, c, d, a, d) and output order $(5, 2, 3, 6, 1, 4)$. The cubic sums are (Y, X, W, Z, Y, Z) and the length is $2Z - 2Y + X - W$. Taking $(Y, Z) = (64, 125)$, the choices $(X, W) = (8, 64), (27, 64), (8, 27)$ give $n = 66, 85, 103$, respectively.

Family IV (twelve blocks and three binary switches). Let X, T, Q be integers and put

$$\begin{aligned} u &= X - 7, & v &= T - 2X + 7, & w &= X + 27 - T, \\ z &= Q + 2T - 2X - 121, & h &= X + 152 - T - Q. \end{aligned}$$

Assume $u, v, z \geq 0$ and $w, h > 0$. Use input lengths

$$(3, u, v, w, 3, w, z, h, 3, 27, 3, h)$$

and output block order

$$(11, 2, 3, 6, 9, 4, 7, 12, 5, 10, 1, 8).$$

The reflected-block lemma computes the twelve nominal index-image sums as

$$(125, X, T, 64, 125, 64, Q, 216, 125, 216, 125, 216).$$

When the second, third, or seventh block is nonempty, its nominal sum X, T, or Q, respectively, must be a cube. Any zero-length block is omitted from both orders and imposes no cube condition. The total length is

$$n = 276 + X - T - Q.$$

Take $X \in \{7, 8\}$, choose $T \in \{2X - 7, 27\}$, and independently choose $Q \in \{121 - 2T + 2X, 125\}$. All eight resulting length choices satisfy the positivity and cube conditions. The 2^3 distinct lengths are

$$\boxed{131, 132, 150, 151, 155, 156, 174, 175}.$$

This is one symbolic rearrangement construction with three optional empty blocks, not eight separate finite endpoint certificates.

Family V (15 blocks, two constrained switches). Take $\alpha, \beta \in \{0, 1\}$ with $\alpha \leq \beta$ and $t \in \{0, 7 + 4\beta + 2\alpha\}$. Define the ordered input block lengths

$$\begin{gathered} \ell_1 = 2\beta, \quad \ell_2 = 7 - \alpha - 4\beta - \alpha\beta, \quad \ell_3 = \alpha, \quad \ell_4 = \alpha\beta, \\ \ell_5 = 13 + 4\beta, \quad \ell_6 = \alpha, \quad \ell_7 = 23 - 4\beta - \alpha, \quad \ell_8 = 13 + 4\beta, \\ \ell_9 = 48 - \alpha - 45\beta + \alpha\beta, \quad \ell_{10} = 2\beta, \quad \ell_{11} = \beta(39 - \alpha), \\ \ell_{12} = t, \quad \ell_{13} = 30 + \alpha - t, \quad \ell_{14} = 13 + 4\beta, \quad \ell_{15} = 91 + \alpha - t. \end{gathered}$$

The output block order is

$$(10, 4, 6, 2, 8, 3, 11, 1, 9, 14, 7, 12, 15, 5, 13).$$

All lengths are nonnegative, with empty blocks omitted. Because $\alpha\beta = \alpha$, the nominal reflected sums are

$$\begin{gathered} 21 + 43\beta,\ 8,\ 28 - \alpha,\ 8,\ 216, \\ 21 + 2\alpha + 4\beta,\ 125,\ 64,\ 125,\ 105 - 41\beta, \\ 125,\ 209 - 2\alpha - 4\beta + t,\ 343,\ 216,\ 343. \end{gathered}$$

Every *nonempty* block has a cubic sum: blocks $1, 10, 11$ are used only if $\beta = 1$; blocks $3, 4, 6$ are used only if $\alpha = \beta = 1$; and block 12 is empty if $t = 0$, otherwise its sum is 216. The total length is

$$\boxed{n = 238 + \alpha + 2\beta - t},$$

so the three allowed pairs $(\alpha, \beta) = (0, 0), (0, 1), (1, 1)$ with their two possible t values give

$$\boxed{228, 229, 231, 238, 240, 241}.$$

This is a symbolic cubic-sum construction; it does not require verifying six separate endpoint lists.

Lemma 4 (Affine-word rigidity)**.** *Fix a finite list of integer cubes $C_1, \ldots, C_t$. Suppose all breakpoints $b_j(n)$ are affine functions of n with $b_0(n) = 0$ and $b_t(n) = n$. The resulting fixed-cube reflection word can be a valid permutation for at most two distinct values of n.*

Proof. Every legal permutation satisfies the sum identity

$$\sum_{j=1}^{t} C_j\big(b_j(n) - b_{j-1}(n)\big) = \sum_{i=1}^{n}(i + p(i)) = n(n+1).$$

Its left side is affine in n; its right side is a monic quadratic. The equality cannot hold for three distinct values. □

The obstruction concerns *fixed* cube sums and affine endpoints only, not varying cube sums or composition of reflection templates.

In total these *five symbolic identities* give twenty-two base lengths

$$66, 85, 100, 103, 119, 122, 131, 132, 150, 151, 155, 156, 174, 175, 182, 228, 229, 231, 238, 240, 241, 254$$

without twenty-two separate endpoint-table checks.

The remaining 60 rows. The appendix records 60 exceptional rows of the form $n\ b_1 : k_1\ \cdots\ b_t : k_t$. The corresponding cubical sums are $c_j = k_j^3$ and the first boundary is $b_0 = 0$. For every row, the boundaries increase, $1 \le L_j < R_j \le n + 1$, and the multiset equality in Lemma 1 holds exactly. These are 816 simple integer-interval identities. Exactly 10 listed rows have $n < 173$ and exactly 50 have $173 \le n \le 364$. Together with the twenty-two algebraic-template lengths, there are 22 lower and 60 upper seeds.

The following two lists give every omitted base length without further matching data. For any tabulated predecessor m under cube c, define $n = c - m - 1$ and apply Lemma 2. The lists are ordered by cube, so every predecessor is either a seed or was obtained in an earlier row.

First: 26 reflected lengths below 173.

```
cube     already constructed predecessors m
  8      0
 27      0,7
 64      0,7,19,26
125      0,7,19,26,37,44,56
216      44,56,63,66,68,80,82,85,87,
         100,103,105
```

Together with the 10 exceptional lower seeds and the twelve algebraic-template lower seeds, these supply 48 distinct lengths below 173.

Second: 49 reflected lengths in $[173, 255] \cup [339, 364]$.

```
cube     previously constructed predecessors m
216      0,7,19,26,37
343      0,87,98,100,103,105,110,112,
         115,116,117,118,119,120,121,122,
         123,124,125,126,128,129,130,131,
         132,133,135,147,149,150,151,152,
         155,156,159
512      147,149,150,151,152,155,156,
         159,171
```

The 50 upper exceptional rows, the ten upper algebraic-template lengths and these 49 reflections are exactly the 109 integers in $[173, 255] \cup [339, 364]$. There is no circular dependency: each predecessor m is strictly smaller than its constructed length n.

For example, $211 = 343 - 131 - 1$, $340 = 512 - 171 - 1$, and $342 = 343 - 0 - 1$. The first example replaces a previously necessary-looking 53-segment independent row by one suffix reflection.

Lemma 5 (Complete finite interval)**.** *For every integer $173 \le n \le 364$, a cube-sum permutation of $[1, n]$ exists.*

Proof. The explicit arithmetic rows and the two predecessor tables establish the claim for $[173, 255] \cup [339, 364]$. For each of the 83 remaining lengths $256 \le n \le 338$, take $m = 511 - n \in [173, 255]$. Then $m < n$ and $m + n + 1 = 512 = 8^3$, so Lemma 2 applies. □

3 A purely symbolic infinite tail

Theorem 1. *For every integer $n \ge 173$, there is a permutation p of $[1, n]$ such that $i + p(i)$ is a perfect cube for all indices i.*

Proof. The base interval is Lemma 5. For $n \ge 365$, assume the claim for every smaller integer at least 173, and let k^3 be the least integer cube satisfying $k^3 \ge n + 174$. Thus $m = k^3 - n - 1 \ge 173$.

If $365 \le n \le 555$, then $k = 9$ and $k^3 = 729 \le 2n$. For $n \ge 556$, minimality of k implies $k \ge 10$ and $n \ge (k-1)^3 - 173$. Therefore

$$\begin{aligned} 2n - k^3 &\ge 2(k-1)^3 - 346 - k^3 \\ &= k^3 - 6k^2 + 6k - 348 \\ &\ge 112 > 0. \end{aligned}$$

The final polynomial equals 112 at $k = 10$ and increases thereafter: its first difference is $3k^2 - 9k + 1 > 0$. In either case $k^3 \le 2n$ and hence $m < n$. Apply the induction hypothesis at m and Lemma 2. □

Logical scope. The theorem uses explicit finite arithmetic equalities and short symbolic lemmas. The optional script `verify_60_plus_templates.py` checks the 60 exceptional rows, regenerates all twenty-two template cases and checks finite closure without matching or other external data. It is not a premise for the proof. A uniform, case-free formula replacing the remaining 60 exceptional rows is *not* claimed.

A The 60 exceptional integer-multiset witnesses

Every token $b : k$ means $p(i) = k^3 - i$ up to endpoint b. For each row, verify the same endpoint balance of Lemma 1.

82: 2:4 5:2 21:3 42:4 61:5 63:4 82:5

116: 2:4 3:3 4:2 8:3 9:5 10:3 11:5 15:3 16:5 17:3 18:5 22:3 23:5 24:3 61:5 63:4 100:5 101:6 102:5 106:6 107:5 108:6 109:5 113:6 114:5 115:6 116:5

118: 1:3 2:2 3:3 4:2 5:3 6:2 7:5 8:3 9:5 10:3 11:5 12:3 14:5 15:3 16:5 17:3 18:5 19:3 21:5 22:3 23:5 24:3 25:5 26:3 98:5 99:6 100:5 101:6 102:5 103:6 105:5 106:6 107:5 108:6 109:5 110:6 112:5 113:6 114:5 115:6 116:5 117:6 118:5

120: 2:4 3:3 4:2 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 120:5

121: 2:4 3:3 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 121:5

123: 2:4 31:5 32:4 61:5 63:4 92:5 93:6 122:5 123:6

125: 3:5 4:2 29:5 30:4 33:5 34:4 90:5 91:6 94:5 95:6 120:5 121:6 124:5 125:6

126: 2:4 5:2 28:5 35:4 61:5 63:4 89:5 96:6 119:5 126:6

127: 1:5 4:2 5:3 6:2 11:5 12:3 18:5 19:3 25:5 26:3 27:5 30:4 31:5 36:4 88:5 91:6 92:5 97:6 102:5 103:6 109:5 110:6 116:5 117:6 118:5 121:6 122:5 127:6

129: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 86:5 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 129:6

173: 7:2 12:3 14:5 19:3 21:5 42:4 103:6 105:5 110:6 112:5 173:6

176: 7:2 12:3 14:5 17:4 19:3 24:5 42:4 103:6 105:5 110:6 115:5 166:6 169:7 176:6

177: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 177:6

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180: 5:3 7:5 12:3 14:5 19:3 21:5 26:3 28:5 35:4 96:6 98:5 103:6 105:5 110:6 112:5 117:6 119:5 180:6

181: 3:5 4:2 29:5 30:4 33:5 34:4 91:6 94:5 95:6 120:5 121:6 124:5 181:6

184: 7:2 31:4 32:6 56:4 159:6 183:7 184:6

185: 21:5 26:3 30:4 51:6 58:4 91:6 112:5 178:6 185:7

188: 2:5 5:4 7:2 30:4 32:6 61:4 91:6 93:5 154:6 157:7 159:6 188:7

194: 7:2 12:3 14:5 19:3 21:5 103:6 105:5 110:6 112:5 194:6

197: 8:5 18:3 55:6 63:4 91:6 99:5 182:6 190:7 197:6

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